Versión de Prueba
Wondershare PDFelement

1) $H(z) = \dfrac{z^{-2} - \alpha}{(1 - \frac{1}{2}z^{-1})(1 - 3z^{-1})}$ α: Nro desconocido.

$x[n] = 2^n \rightarrow y[n] = 0 \ \forall n$

Entonces. $0 = H(z) \cdot 2^n \Rightarrow H(z) = 0$

$H(z) = \dfrac{z^{-2} - \alpha}{(1 - \frac{1}{2}\cdot 2^{-1})(1 - 3\cdot 2^{-1})} = 0$

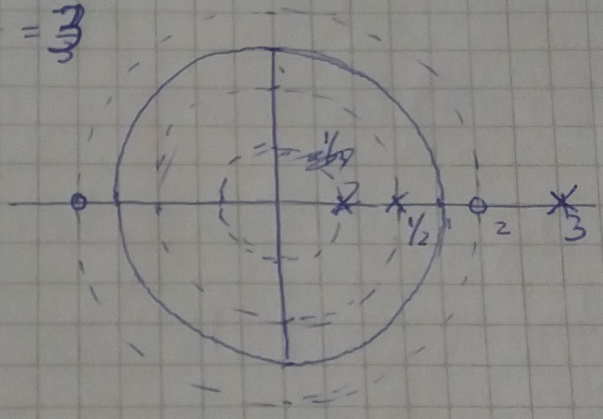
$\Rightarrow \dfrac{1/4 - \alpha}{(1 - \frac{1}{4})(1 - \frac{3}{2})} = 0 \Rightarrow \dfrac{1/4 - \alpha}{3/4 \cdot (-1/2)} = 0 \Rightarrow \boxed{\alpha = 1/4}$

Entonces: $H(z) = \dfrac{z^{-2} - 1/4}{(1 - \frac{1}{2}z^{-1})(1 - 3z^{-1})}$

Polos: p_1: $1 = \frac{1}{2z} \Rightarrow z = \frac{1}{2}$

p_2: $1 = \frac{3}{z} \Rightarrow z = 3$

Ceros: $1/4 = \frac{1}{z^2} \Rightarrow z^2 = 4 \Rightarrow \boxed{z = \pm 2}$



ROC causal y estable: no existe.

ROC causal: $|z| > 3$

ROC estable: $|z| < 3$ (anticausal)

c) $y[n] = H(z) \cdot z^n$ / $x[n] = 2^n$. Si $x[n] = 1$, $y[n] = H(1) \cdot 1^n$

$H(1) = \dfrac{1^{-2} - 1/4}{(1 - 1/2)(1 - 3)} = \dfrac{3/4}{(1/2)(-2)} = \dfrac{3/4}{-1/1} = -\dfrac{3}{4}$

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15 - 12 - 14

a) $Re[x[n]] \Rightarrow X_e[n] = \frac{x[n] + x[-n]}{2}$

La DFT de $X[n]$ se define como:

$$X[\Omega] = \sum_{n=0}^{N-1} x[n] e^{-j\Omega n} \longrightarrow X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

$$Re[X[k]] = Re\left[\sum_{n=0}^{N-1} x[n] \left(\cos\left[\frac{2\pi}{N}kn\right] - j\,sen\left[\frac{2\pi}{N}kn\right] \right) \right]$$

$$= \sum_{n=0}^{N-1} x[n] \cos\left[\frac{2\pi}{N}kn\right] \quad (A)$$

Por otro lado:

$x[-n] \xrightarrow{DFT} X(e^{j\Omega})$

$$DFT[X_e[n]] = \sum_{n=0}^{N-1} \left[\frac{x[n]}{2} e^{-j\frac{2k\pi n}{N}} + \frac{x[n]}{2} e^{j\frac{2k\pi n}{N}} \right]$$

$$= \sum_{n=0}^{N-1} \left[\frac{x[n]}{2} \left(\cos\left[\frac{2k\pi n}{N}\right] - j\,sen\left[\frac{2k\pi n}{N}\right] + \cos\left[\frac{2\pi kn}{N}\right] + j\,sen\left[\frac{2\pi kn}{N}\right] \right) \right]$$

$$= \sum_{n=0}^{N-1} \frac{x[n]}{2} \left(2\cos\left[\frac{2k\pi n}{N}\right] \right) = \sum_{n=0}^{N-1} x[n] \cos\left[\frac{2k\pi n}{N}\right] \quad (B)$$

(A) = (B). $Re(X[k]) = DFT[X_e[n]]$

b) Es la parte par de $X[n]$, es decir $X_e[n] = \frac{x[n] + x[-n]}{2}$